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Muhammad Adnan

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ABOUT MYSELF

I hold a PhD in Interactive and Cognitive Environments from a joint programme between the University of Genoa and Universidad Carlos III de Madrid, where I focused on LiDAR-based perception and vehicle localization using probabilistic models. I have over five years of experience in teaching, software development, and research, with strong interests in machine learning, computer vision, and autonomous systems.

EDUCATION & TRAINING

01/01/2022 - 16/12/2025 - GENOVA, ITALY

DOCTOR OF PHILOSOPHY (PHD) IN INTERACTIVE AND COGNITIVE ENVIRONMENTS- UNIVERSITY OF GENOVA

Level in EQF: 8

01/01/2008 - 25/02/2010 - KARLSKRONA, SWEDEN

MASTER OF SCIENCE IN COMPUTER SCIENCE (MSC)- BLEKINGE INSTITUTE OF TECHNOLOGY

Level in EQF: 7

01/01/2002 - 07/07/2007 - PESHAWAR, PAKISTAN

BACHELOR OF SCIENCE IN COMPUTER SCIENCE (BSC HONS)- UNIVERSITY OF PESHAWAR

Level in EQF: 6

WORK EXPERIENCE

01/10/2025 - CURRENT - RENDE, ITALY

COMPUTER SCIENTIST DEPARTMENT OF COMPUTER, MODELING, ELECTRONIC AND SYSTEMS ENGINEERING, UNIVERSITY OF CALABRIA

- Appointed under the **Italian National Recovery and Resilience Plan (PNRR)**, Mission 4, Component 2, Investment 1.3, funded by the **European Union – NextGenerationEU**.

Role: Computer Science Expert – Large Language Models (LLMs) and Knowledge Graphs.

- Contributing to the **design and development of explainable, knowledge-grounded AI systems**, integrating LLMs with knowledge-graph reasoning for transparent and interpretable decision-making.

- Actively involved in research aligned with the **FAIR project**, focusing on **trustworthy, explainable, and interoperable AI technologies**.

01/01/2022 - 16/12/2025 - GENOVA, ITALY

EDUCATIONAL RESEARCHER UNIVERSITY OF GENOVA

- **PhD Researcher on a fully funded doctoral scholarship**, conducting advanced research on **LiDAR point cloud processing** for object tracking and ego-vehicle localization in autonomous driving scenarios.

- Designed and implemented **probabilistic classification models for static and dynamic objects**, leveraging **Dynamic Bayesian Networks (DBNs)** and **generative modeling approaches** for robust perception under

uncertainty.

- Successfully completed **advanced doctoral coursework** in **Artificial Intelligence, Machine Learning, Deep Learning, and probabilistic modeling**, with a strong theoretical and applied foundation.
- Authored peer-reviewed scientific publications in **Sensors**, **Applied Sciences**, and leading international venues including **ICFSP 2024** and **IJCNN 2025**.
- Presented research outcomes at **international conferences** and actively participated in **AI-focused workshops and summer schools**, contributing to knowledge exchange and interdisciplinary collaboration.

04/04/2016 - 29/12/2021 - SWABI, PAKISTAN

COMPUTER SCIENCE LECTURER UNIVERSITY OF SWABI

- Delivered undergraduate courses in the computer science department with a focus on ML, DL, and web development.
- Designed curricula, supervised lab sessions, and assessed student performance.
- Conducted independent research and presented at national and international conferences.
- Organising the academic seminars and contributing to international event coordination.
- Collaborated with department leadership to support student success and curriculum development.

01/01/2013 - 28/02/2016 - PESHAWAR, PAKISTAN

WEB DEVELOPER KHYBER PAKHTUNKHWA OIL AND GAS COMPANY LIMITED

- Designed, developed, and maintained the company's official website with a focus on usability and performance.
- Implemented responsive layouts, integrated APIs, and optimized loading times across platforms.
- Ensured data security and website reliability through regular updates and issue resolution.
- Aligned web solutions with business goals through collaboration with stakeholders.
- Provided staff training and maintained documentation to support ongoing web operations.

LANGUAGE SKILLS

Mother tongue(s): **URDU**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2
ITALIAN	A1	A1	A1	A1	A1

Levels: A1 and A2: Basic user - B1 and B2: Independent user - C1 and C2: Proficient user

SKILLS

web programming | MATLAB | COMPUTER VISION | Python (computer programming) | Python HuggingFace

PUBLICATIONS

A generative model approach for LiDAR-based classification

Adnan, Muhammad, et al. "A Generative Model Approach for LiDAR-Based Classification and Ego Vehicle Localization Using Dynamic Bayesian Networks." Applied Sciences 15.9 (2025): 5181.

Systematic and Comprehensive Review of Clustering and Multi-Target Tracking Techniques

Adnan, Muhammad, et al. "Systematic and comprehensive review of clustering and multi-target tracking

techniques for LiDAR point clouds in autonomous driving applications." Sensors 23.13 (2023): 6119.

Autonomous Vehicle Localization via LiDAR-Based Classification of Dynamic and Static Tracks

Adnan, Muhammad, et al. "Autonomous Vehicle Localization via LiDAR-Based Classification of Dynamic and Static Tracks in Dynamic Environments." 2025 International Joint Conference on Neural Networks (IJCNN). IEEE, 2025. International joint conference on neural networks

Classifying Static and Dynamic Tracks for LiDAR-Based Navigation of Autonomous Vehicle Systems

Adnan, Muhammad, et al. "Classifying Static and Dynamic Tracks for LiDAR-Based Navigation of Autonomous Vehicle Systems." 2024 9th International Conference on Frontiers of Signal Processing (ICFSP). IEEE, 2024.

Towards EnergyEfficient Framework for IoT Big Data Healthcare Solutions

Feng, Chong, et al. "Towards EnergyEfficient Framework for IoT Big Data Healthcare Solutions." Scientific Programming 2020.1 (2020): 7063681.

Towards an Improved Energy Efficient and EndtoEnd Secure Protocol for IoT Healthcare Applications

Ahmad, Arshad, et al. "Towards an Improved Energy Efficient and EndtoEnd Secure Protocol for IoT Healthcare Applications." Security and Communication Networks 2020.1 (2020): 8867792.

Smart Pilot Decontamination Strategy for High and Low Contaminated Users in Massive MIMO-5G Network

Khan, Khalid, et al. "Smart Pilot Decontamination Strategy for High and Low Contaminated Users in Massive MIMO-5G Network." ICITA 2022. Singapore: Springer Nature Singapore, 2023.

CERTIFICATIONS

01/07/2022 MaLGa (University of Genova)
Machine Learning Crash Course 2022

20/07/2022 MaLGa (University of Genova)
Deep Learning: a hands-on introduction 2022

20/07/2022 MaLGa (University of Genova)
Computer Vision Crash Course 2022

18/04/2022 Interactive and cognitive environments (JDICE), University of Genova
Probabilistic reasoning/Dynamic Bayesian networks

04/07/2024 University of Udine
International summer school on Artificial Intelligence 2023

30/07/2024 University of Genova, University Carlos III
Summer course on autonomous systems

CONFERENCES & SEMINARS

27/11/2018 - 28/11/2018 Vancouver, Canada
IoT, Big Data Healthcare Summit Western Canada

02/07/2023 - 03/07/2023 University of Udine, Italy
Generative AI, Stable Diffusion, and the Revolution in Visual Synthesis

12/09/2024 - 14/09/2024 Paris
International Conference on Frontiers of Signal Processing (ICFSP)